

AMENDMENT NO. 1 DECEMBER 2025
TO
IS 737 : 2024 WROUGHT ALUMINIUM AND ALUMINIUM ALLOY SHEET AND
STRIP FOR GENERAL ENGINEERING PURPOSES — SPECIFICATION
(Fifth Revision)

(Page 1, clause 5) — Insert the following at the end:

‘The aluminium and aluminium alloy sheet and strip may be supplied in lacquered, painted or any other coating. The requirement of coating shall be as agreed between the purchaser and the manufacturer, however, the base material shall conform to the requirements of **7, 8 and 10** of IS 737.’

(Page 1, clause 8.1) — Substitute the following for the existing:

‘**8.1** The material when tested in accordance with IS 1608 (Part 1) shall conform to the values given in Table 2 for grades mentioned in Table 1. The mechanical properties of the grades, conditions or temper designations and thickness not mentioned in Table 2, may be subject to the mutual agreement between the purchaser and the manufacturer.

The tensile test piece shall be rectangular section and conform to the dimension as given in IS 1608 (Part 1) with a gauge length of 50 mm. For sheet and strip of width 300 mm and above, the test piece shall be taken in a direction transverse to that of rolling. For sheet and strip of width less than 300 mm, the test piece shall be taken in a direction parallel to that of rolling. When the width of the material to be tested is insufficient to permit preparation of the standard tensile test piece, a piece of the full width of the material may be used.’

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(Page 3, clause 7, Table 1) — Substitute the following for the existing table:

Table 1 Chemical Composition of Wrought Aluminium and Aluminium Alloy Sheet and Strip
(Composition Limits are in Weight Percent Maximum, Unless Shown Otherwise)

(Clauses 7 and 8.1)

Sl No.	IS Designation	ISO Designation	Si	Fe	Cu	Mn	Mg	Cr	Ni	Zn	Ti	Others		Al	Remarks
												Each	Total		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
1)	19000	—	0.50	0.70	0.10	0.10	0.20	—	—	0.10	—	—	—	99.00, <i>Min</i>	Cu + Mg + Si + Fe + Mn + Zn : 1.00
2)	19000A	1200	—	—	0.05	0.05	—	—	—	0.10	0.05	0.05	0.15	99.00, <i>Min</i>	Si + Fe : 1.00
3)	19000B	1200A	—	—	0.10	0.30	0.30	0.10	—	0.10	—	0.05	0.15	99.00, <i>Min</i>	Si + Fe : 1.00
4)	19002	1100	—	—	0.05 to 0.20	0.05	—	—	—	0.10	—	0.05	0.15	99.00, <i>Min</i>	Si + Fe : 0.95
5)	19350	1235	—	—	0.05	0.05	0.05	—	—	0.10	0.06	0.03	—	99.35, <i>Min</i>	Si + Fe : 0.65, V : 0.05
6)	19450	1145	—	—	0.05	0.05	0.05	—	—	0.05	0.03	0.03	—	99.45, <i>Min</i>	Si + Fe : 0.55, V : 0.05
7)	19500	1050	0.25	0.40	0.05	0.05	0.05	—	—	0.05	0.03	0.03	—	99.50, <i>Min</i>	V : 0.05
8)	19500A	1050A	0.25	0.40	0.05	0.05	0.05	—	—	0.07	0.05	0.03	—	99.50, <i>Min</i>	—
9)	19501	—	0.15	0.35	0.04	0.02	—	0.02	—	0.02	—	0.03	0.10	99.50, <i>Min</i>	Ti + V : 0.02
10)	19501A	1350	0.10	0.40	0.05	0.01	—	0.01	—	0.05	—	0.03	0.10	99.50 <i>Min</i>	Ti + V : 0.02, B : 0.05, Ga : 0.03
11)	19600	1060	0.25	0.35	0.05	0.03	0.03	—	—	0.05	0.03	0.03	—	99.60, <i>Min</i>	V : 0.05
12)	19700	1070	0.20	0.25	0.04	0.03	0.03	—	—	0.04	0.03	0.03	—	99.70, <i>Min</i>	V : 0.05
13)	19700A	1070A	0.20	0.25	0.03	0.03	0.03	—	—	0.07	0.03	0.03	—	99.70, <i>Min</i>	—
14)	19800	1080	0.15	0.15	0.03	0.02	0.02	—	—	0.03	0.03	0.02	—	99.80, <i>Min</i>	Ga : 0.03, V : 0.05

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												Each	Total		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
15)	19800A	1080A	0.15	0.15	0.03	0.02	0.02	–	–	0.06	0.02	0.02	–	99.80, <i>Min</i>	Ga: 0.03
16)	19850	1085	0.10	0.12	0.03	0.02	0.02	–	–	0.03	0.02	0.01	–	99.85, <i>Min</i>	Ga : 0.03, V : 0.05
17)	19900	1190	0.05	0.07	0.01	0.01	0.01	0.01	–	0.02	–	0.01	–	99.90, <i>Min</i>	V + Ti : 0.01, B : 0.01, Ga : 0.02
18)	19910	1110	0.30	0.80	0.04	0.01	0.25	0.01	–	–	–	0.03	–	99.10, <i>Min</i>	V + Ti : 0.03, B: 0.02
19)	19950	1350A	0.25	0.40	0.02	–	0.05	–	–	0.05	–	0.03	–	99.50, <i>Min</i>	Cr + Mn + Ti + V: 0.03
20)	19950A	1450	0.25	0.40	0.05	0.05	0.05	–	–	0.07	0.10 to 0.20	0.03	–	99.50, <i>Min</i>	–
21)	19970	1370	0.10	0.25	0.02	0.01	0.02	0.01	–	0.04	–	0.02	0.10	99.70, <i>Min</i>	V + Ti : 0.02, B : 0.02, Ga : 0.03
22)	19990	1199	0.006	0.006	0.006	0.002	0.006	–	–	0.006	0.002	0.002	–	99.99, <i>Min</i>	Ga : 0.005, V : 0.005
23)	19990A	1090	0.07	0.07	0.02	0.01	0.01	–	–	0.03	0.01	0.01	–	99.90, <i>Min</i>	Ga : 0.03, V:0.05
24)	19998	1198	0.01	0.006	0.006	0.006	–	–	–	0.01	0.006	0.003	–	99.98, <i>Min</i>	Ga: 0.006
25)	22130	2091	0.20	0.30	1.80 to 2.50	0.10	1.10 to 1.90	0.10	–	0.25	0.10	0.05	0.15	Remainder	Li: 1.70 to 2.30; Zr: 0.04 to 0.16
26)	22570	2031	0.50 to 1.30	0.60 to 1.20	1.80 to 2.80	0.50	0.60 to 1.20	–	0.60 to 1.40	0.20	0.20	0.05	0.15	Remainder	–
27)	22580	2618A	0.15 to 0.25	0.90 to 1.40	1.80 to 2.70	0.25	1.20 to 1.80	–	0.80 to 1.40	0.15	0.20	0.05	0.15	Remainder	Ni : 0.80 to 1.40, Zr + Ti : 0.25
28)	23540	2001	0.20	0.20	5.20 to 6.00	0.15 to 0.50	0.20 to 0.45	0.10	0.05	0.10	0.20	0.05	0.15	Remainder	Pb : 0.003; Zr : 0.05
29)	23660	2214	0.50 to 1.20	0.30	3.90 to 5.00	0.40 to 1.20	0.20 to 0.80	0.10	–	0.25	0.15	0.05	0.15	Remainder	–

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												Each	Total		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
30)	24345	–	0.50 to 1.20	0.70	3.80 to 5.00	0.30 to 1.20	0.20 to 0.80	0.30	–	0.20	0.30	–	–	Remainder	Ti + Cr : 0.30
31)	24345A	2014	0.50 to 1.20	0.70	3.90 to 5.00	0.40 to 1.20	0.20 to 0.80	0.10	–	0.25	0.15	0.05	0.15	Remainder	–
32)	24345B	2014A	0.50 to 0.90	0.50	3.90 to 5.00	0.40 to 1.20	0.20 to 0.80	0.10	0.10	0.25	0.15	0.05	0.15	Remainder	Zr + Ti : 0.20
33)	24530	2024	0.50	0.50	3.80 to 4.90	0.30 to 0.90	1.20 to 1.80	0.10	–	0.25	0.15	0.05	0.15	Remainder	–
34)	24534	–	0.20 to 0.70	0.70	3.50 to 4.70	0.40 to 1.20	0.40 to 1.20	–	–	0.20	0.30	–	–	Remainder	–
35)	24534A	2017A	0.20 to 0.80	0.70	3.50 to 4.50	0.40 to 1.00	0.40 to 1.00	0.10	–	0.25	–	0.05	0.15	Remainder	–
36)	24534B	2017	0.20 to 0.80	0.70	3.50 to 4.50	0.40 to 1.00	0.40 to 0.80	0.10	–	0.25	0.15	0.05	0.15	Remainder	–
37)	26388	2219	0.20	0.30	5.80 to 6.80	0.20 to 0.40	0.02	–	–	0.10	0.02 to 0.10	0.05	0.15	Remainder	V : 0.05 to 1.50, Zr : 0.10 to 0.25
38)	26388A	2319	0.20	0.30	5.80 to 6.80	0.20 to 0.40	0.02	–	–	0.10	0.10 to 0.20	0.05	0.15	Remainder	Zr : 0.10 to 0.25, V : 0.05 to 0.15
39)	31000	–	0.60	0.70	0.10	0.80 to 1.50	0.10	0.20	–	0.20	0.20	–	–	Remainder	–
40)	31000A	3103	0.50	0.70	0.10	0.90 to 1.50	0.30	0.10	–	0.20	–	0.05	0.15	Remainder	Zr + Ti : 0.10

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												Each	Total		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
41)	31200	3003	0.60	0.70	0.05 to 0.20	1.00 to 1.50	—	—	—	0.10	—	0.05	0.15	Remainder	—
42)	31300	3002	0.08	0.10	0.15	0.05 to 0.25	0.05 to 0.20	—	—	0.05	0.03	0.03	0.10	Remainder	V : 0.05
43)	31400	3102	0.40	0.70	0.10	0.05 to 0.40	—	—	—	0.25	0.10	0.05	0.15	Remainder	—
44)	31440	3017	0.25	0.25 to 0.45	0.25 to 0.40	0.80 to 1.20	0.10	0.15	—	0.10	0.05	0.05	0.15	Remainder	—
45)	31500	—	0.40	0.70	0.20	1.00 to 1.50	0.60 to 1.30	—	—	0.20	0.20	—	—	Remainder	—
46)	31500A	3004	0.30	0.70	0.25	1.00 to 1.50	0.80 to 1.30	—	—	0.25	—	0.05	0.15	Remainder	—
47)	31500B	—	0.40	0.70	0.10	0.30 to 0.80	0.20 to 0.80	0.10	—	0.20	—	—	—	Remainder	—
48)	31500C	3105A	0.60	0.70	0.30	0.30 to 0.80	0.20 to 0.80	0.20	—	0.25	0.10	0.05	0.15	Remainder	—
49)	31540	3104	0.60	0.80	0.05 to 0.25	0.80 to 1.40	0.80 to 1.30	—	—	0.25	0.10	0.05	0.15	Remainder	V : 0.05, Ga : 0.05
50)	31542	3005	0.60	0.70	0.30	1.00 to 1.50	0.20 to 0.60	0.10	—	0.25	0.10	0.05	0.15	Remainder	—
51)	31542A	3005A	0.70	0.80	0.30	1.00 to 1.50	0.20 to 0.60	0.10	—	0.40	0.10	0.05	0.15	Remainder	—

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												Each	Total		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
52)	31600	3105B	0.70	0.90	0.30	0.30 to 0.90	0.20 to 0.80	0.20	–	0.50	0.10	0.05	0.15	Remainder	Pb : 0.10
53)	32400	3207	0.30	0.45	0.10	0.40 to 0.80	0.10	–	–	0.10	–	0.05	0.10	Remainder	–
54)	32400A	3207A	0.35	0.60	0.25	0.30 to 0.80	0.40	0.20	–	0.25	–	0.05	0.15	Remainder	–
55)	40800	–	0.60 to 0.95	0.60 to 0.95	0.20	0.10	0.10	–	–	0.20	0.20	–	–	98.0, <i>Min</i>	–
56)	41352	4015	1.40 to 2.20	0.70	0.20	0.60 to 1.20	0.10 to 0.50	–	–	0.20	–	0.05	0.15	Remainder	–
57)	41352A	4017	0.60 to 1.60	0.70	0.10 to 0.50	0.60 to 1.20	0.10 to 0.50	–	–	0.20	–	0.05	0.15	Remainder	–
58)	41352B	4115	1.80 to 2.20	0.70	0.10 to 0.50	0.60 to 1.20	0.10 to 0.50	–	–	0.20	–	0.05	0.15	Remainder	–
59)	41354	4016	1.40 to 2.20	0.70	0.20	0.60 to 1.20	0.10	–	–	0.50 to 1.30	–	0.05	0.15	Remainder	–
60)	41800	4006	0.80 to 1.20	0.50 to 0.80	0.10	0.05	0.01	0.20	–	0.05	–	0.05	0.15	Remainder	–
61)	43000	–	4.50 to 6.00	0.60	0.10	0.50	0.20	–	–	0.20	–	–	–	Remainder	–
62)	43000A	4043	4.50 to 6.00	0.80	0.30	0.05	0.05	–	–	0.10	0.20	0.05	0.15	Remainder	–

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												Each	Total		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
63)	43000B	4043A	4.50 to 6.00	0.60	0.30	0.15	0.20	—	—	0.10	0.15	0.05	0.15	Remainder	—
64)	44000	4343	6.80 to 8.20	0.80	0.25	0.10	—	—	—	0.20	—	0.05	0.15	Remainder	—
65)	44800	4018	6.50 to 7.50	0.20	0.05	0.10	0.50 to 0.80	—	—	0.10	0.20	0.05	0.15	Remainder	—
66)	45000	4045	9.00 to 11.0	0.80	0.30	0.05	0.05	—	—	0.10	0.20	0.05	0.15	Remainder	—
67)	45500	4046	9.00 to 11.0	0.50	0.03	0.40	0.20 to 0.50	—	—	0.10	0.15	0.05	0.15	Remainder	—
68)	45580	4004	9.00 to 10.5	0.80	0.25	0.10	1.00 to 2.00	—	—	0.20	—	0.05	0.15	Remainder	—
69)	45580A	4104	9.00 to 10.5	0.80	0.25	0.10	1.00 to 2.00	—	—	0.20	—	0.05	0.15	Remainder	Bi : 0.02 to 0.20
70)	46000	—	10.0 to 13.0	0.60	0.10	0.50	0.20	—	—	0.20	—	—	—	Remainder	—
71)	46000A	4047	11.0 to 13.0	0.80	0.30	0.15	0.10	—	—	0.20	—	0.05	0.15	Remainder	—
72)	46000B	4047A	11.0 to 13.0	0.60	0.30	0.15	0.10	—	—	0.20	0.15	0.05	0.15	Remainder	—
73)	47000	4032	11.0 to 13.5	1.00	0.50 to 1.30	—	0.80 to 1.30	0.10	0.50 to 1.30	0.25	—	0.05	0.15	Remainder	—

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												Each	Total		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
74)	51000A	5005	0.30	0.70	0.20	0.20	0.50 to 1.10	0.10	–	0.25	–	0.05	0.15	Remainder	–
75)	51000B	5050	0.40	0.70	0.20	0.10	1.10 to 1.80	0.10	–	0.25	–	0.05	0.15	Remainder	–
76)	51080	5110	0.08	0.08	–	0.03	0.30 to 0.60	–	–	0.05	0.02	0.02	–	Remainder	–
77)	51080A	5210	0.06	0.04	–	0.03	0.35 to 0.60	–	–	0.04	0.01	0.01	–	Remainder	–
78)	51080B	5310	0.01	0.008	–	–	0.35 to 0.60	–	–	0.01	0.008	0.003	–	Remainder	Fe + Ti : 0.008
79)	51081	5352	–	–	0.10	0.10	2.20 to 2.80	0.10	–	0.10	0.10	0.05	0.15	Remainder	Si + Fe : 0.45
80)	51300	–	0.60	0.90	0.30	0.20 to 0.70	0.20 to 0.90	0.20	–	0.40	0.20	–	–	Remainder	–
81)	51300A	5010	0.40	0.70	0.25	0.10 to 0.30	0.20 to 0.60	0.15	–	0.30	0.10	0.05	0.15	Remainder	–
82)	51800	5006	0.40	0.80	0.10	0.40 to 0.80	0.80 to 1.30	0.10	–	0.25	0.10	0.05	0.15	Remainder	–
83)	51880	5305	0.08	0.08	–	0.03	0.70 to 1.10	–	–	0.05	0.02	0.02	–	Remainder	–
84)	51880A	5505	0.06	0.04	–	0.03	0.80 to 1.10	–	–	0.04	0.01	0.01	–	Remainder	–

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(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
85)	51880B	5605	0.01	0.008	–	–	0.80 to 1.10	–	–	0.01	0.008	0.003	–	Remainder	Fe + Ti : 0.008
86)	52000	–	0.60	0.70	0.10	0.50	1.70 to 2.60	0.25	–	0.20	0.20	–	–	Remainder	Cr + Mn : 0.50
87)	52000A	5251	0.40	0.50	0.15	0.10 to 0.50	1.70 to 2.40	0.15	–	0.15	0.15	0.05	0.15	Remainder	–
88)	52300	–	0.80	0.90	0.20	0.50 to 1.00	1.50 to 2.40	0.20	–	0.40	0.20	–	–	Remainder	–
89)	52300A	–	0.60	0.40 to 0.70	0.20	1.10 to 1.50	1.30 to 1.70	0.20	–	0.40	0.20	–	–	Remainder	–
90)	52302	5049	0.40	0.50	0.10	0.50 to 1.10	1.60 to 2.50	0.30	–	0.20	0.10	0.05	0.15	Remainder	–
91)	52302A	5149	0.25	0.40	0.05	0.50 to 1.10	1.60 to 2.50	0.30	–	0.20	0.15	0.05	0.15	Remainder	–
92)	52302B	5249	0.25	0.40	0.05	0.50 to 1.10	1.60 to 2.50	0.30	–	0.20	0.15	0.05	0.15	Remainder	Zr : 0.10 to 0.20
93)	52700	5050A	0.40	0.70	0.20	0.30	1.10 to 1.80	0.10	–	0.25	–	0.05	0.15	Remainder	–
94)	53000	–	0.60	0.70	0.10	0.50	2.80 to 4.00	0.25	–	0.20	0.20	–	–	Remainder	Cr + Mn : 0.50
95)	53000A	5754	0.40	0.40	0.10	0.50	2.60 to 3.60	0.30	–	0.20	0.15	0.05	0.15	Remainder	Cr + Mn : 0.10 to 0.60

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(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
96)	53380	5454	0.25	0.40	0.10	0.50 to 1.00	2.40 to 3.00	0.05 to 0.20	—	0.25	0.20	0.05	0.15	Remainder	—
97)	53380A	5354	0.25	0.40	0.05	0.50 to 1.00	2.40 to 3.00	0.05 to 0.20	—	0.25	0.15	0.05	0.15	Remainder	Zr : 0.10 to 0.20
98)	53380B	5554	0.25	0.40	0.10	0.50 to 1.00	2.40 to 3.00	0.05 to 0.20	—	0.25	0.05 to 0.20	0.05	0.15	Remainder	—
99)	53600	5018	0.25	0.40	0.05	0.20 to 0.60	2.60 to 3.60	0.30	—	0.20	0.15	0.05	0.15	Remainder	Mn + Cr : 0.20 to 0.60
100)	53640`	5119	0.25	0.40	0.05	0.20 to 0.60	4.50 to 5.60	0.30	—	0.20	0.15	0.05	0.15	Remainder	Mn + Cr : 0.20 to 0.60
101)	53640A	5119A	0.25	0.40	0.05	0.20 to 0.60	4.50 to 5.60	0.30	—	0.20	0.15	0.05	0.15	Remainder	Mn + Cr : 0.20 to 0.60
102)	53800	5052	0.25	0.40	0.10	0.10	2.20 to 2.80	0.15 to 0.35	—	0.10	-	0.05	0.15	Remainder	—
103)	53800A	5252	0.08	0.10	0.10	0.10	2.20 to 2.80	—	—	0.05	—	0.03	0.10	Remainder	V : 0.05
104)	54000	5154A	0.50	0.50	0.10	0.50	3.10 to 3.90	0.25	—	0.20	0.20	0.05	0.15	Remainder	Cr + Mn : 0.10 to 0.50
105)	54000A	5154	0.25	0.40	0.10	0.10	3.10 to 3.90	0.15 to 0.35	—	0.20	0.20	0.05	0.15	Remainder	—
106)	54140	5183	0.40	0.40	0.10	0.50 to 1.00	4.30 to 5.20	0.05 to 0.25	—	0.25	0.15	0.05	0.15	Remainder	—

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SI No.	IS Designation	ISO Designation	Si	Fe	Cu	Mn	Mg	Cr	Ni	Zn	Ti	Others		Al	Remarks
												Each	Total		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
107)	54140A	5183A	0.40	0.40	0.10	0.50 to 1.00	4.30 to 5.20	0.05 to 0.25	–	0.25	0.15	0.05	0.15	Remainder	–
108)	54150	5283A	0.30	0.30	0.03	0.50 to 1.00	4.50 to 5.10	0.05	0.03	0.10	0.03	0.05	0.15	Remainder	Pb : 0.003; Zr : 0.05
109)	54300	–	0.40	0.70	0.10	0.50 to 1.00	4.00 to 4.90	0.25	–	0.20	0.20	–	–	Remainder	–
110)	54300A	5083	0.40	0.40	0.10	0.40 to 1.00	4.00 to 4.90	0.05 to 0.25	–	0.25	0.15	0.05	0.15	Remainder	–
111)	54300B	5182	0.20	0.35	0.15	0.20 to 0.50	4.00 to 5.00	0.10	–	0.25	0.10	0.05	0.15	Remainder	–
112)	54380	5086	0.40	0.50	0.10	0.20 to 0.70	3.50 to 4.50	0.05 to 0.25	–	0.25	0.15	0.05	0.15	Remainder	–
113)	54380A	5186	0.40	0.45	0.25	0.20 to 0.50	3.80 to 4.80	0.15	–	0.40	0.15	0.05	0.15	Remainder	Zr : 0.05
114)	54540	5654	–	–	0.05	0.01	3.10 to 3.90	0.15 to 0.35	–	0.20	0.05 to 0.15	0.05	0.15	Remainder	0.45 Si + Fe
115)	54540A	5654A	–	–	0.05	0.01	3.10 to 3.90	0.15 to 0.35	–	0.20	0.05 to 0.15	0.05	0.15	Remainder	0.45 Si + Fe
116)	55000	–	0.60	0.70	0.10	0.50	4.50 to 5.50	0.25	–	0.20	0.20	0.05	0.15	Remainder	Cr + Mn : 0.50
117)	55000A	5056	0.30	0.40	0.10	0.05 to 0.20	4.50 to 5.60	0.05 to 0.20	–	0.10	–	0.05	0.15	Remainder	–

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SI No.	IS Designation	ISO Designation	Si	Fe	Cu	Mn	Mg	Cr	Ni	Zn	Ti	Others		Al	Remarks
												Each	Total		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
118)	55100	5087	0.25	0.40	0.05	0.70 to 1.10	4.50 to 5.20	0.05 to 0.25	—	0.25	0.15	0.05	0.15	Remainder	Zr : 0.10 to 0.20
119)	55500	5058	0.40	0.50	0.10	0.20	4.50 to 5.60	0.10	—	0.20	0.20	0.05	0.15	Remainder	Pb : 1.20 to 1.80
120)	56400	5356	0.25	0.40	0.10	0.05 to 0.20	4.50 to 5.50	0.05 to 0.20	—	0.10	0.06 to 0.20	0.05	0.15	Remainder	—
121)	56400A	5356A	0.25	0.40	0.10	0.05 to 0.20	4.50 to 5.50	0.05 to 0.20	—	0.10	0.06 to 0.20	0.05	0.15	Remainder	—
122)	55660	5449A	0.60	1.20	0.30	0.60 to 1.10	1.60 to 2.60	0.30	0.10	0.30	0.10	0.05	0.15	Remainder	Sn : 0.10
123)	56420	5456A	0.25	0.40	0.05	0.70 to 1.10	4.50 to 5.20	0.05 to 0.25	—	0.25	0.15	0.05	0.15	Remainder	—
124)	56420A	5456B	0.25	0.40	0.05	0.70 to 1.10	4.50 to 5.20	0.05 to 0.25	—	0.25	0.15	0.05	0.15	Remainder	—
125)	56430	5556A	0.25	0.40	0.10	0.60 to 1.00	5.00 to 5.50	0.05 to 0.20	—	0.20	0.05 to 0.20	0.05	0.15	Remainder	—
126)	56430A	5556B	0.25	0.40	0.10	0.60 to 1.00	5.00 to 5.50	0.05 to 0.20	—	0.20	0.05 to 0.20	0.05	0.15	Remainder	-
127)	57100	5657	0.08	0.10	0.10	0.03	0.60 to 1.00	—	—	0.05	—	0.02	0.05	Remainder	V : 0.05 ; Ga : 0.03
128)	62100	6003	0.35 to 1.00	0.60	0.10	0.80	0.80 to 1.50	0.35	—	0.20	0.10	0.05	0.15	Remainder	—

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SI No.	IS Designation	ISO Designation	Si	Fe	Cu	Mn	Mg	Cr	Ni	Zn	Ti	Others		Al	Remarks
												Each	Total		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
129)	62400	-	0.50 to 0.90	0.70	0.10	0.10	0.40 to 0.60	0.10	—	0.10	0.10	0.05	0.15	Remainder	—
130)	62400A	6005	0.60 to 0.90	0.35	0.10	0.10	0.40 to 0.60	0.10	—	0.10	0.10	0.05	0.15	Remainder	—
131)	62400B	6005B	0.45 to 0.80	0.30	0.10	0.10	0.40 to 0.80	0.10	—	0.10	0.10	0.05	0.15	Remainder	—
132)	62400C	6105	0.60 to 1.00	0.35	0.10	0.15	0.45 to 0.80	0.10	—	0.10	0.10	0.05	0.15	Remainder	—
133)	62550	6201	0.50 to 0.90	0.50	0.10	0.03	0.60 to 0.90	0.03	—	0.10	-	0.03	0.10	Remainder	B : 0.06
134)	63000	6063	0.20 to 0.60	0.35	0.10	0.10	0.45 to 0.90	0.10	—	0.10	0.10	0.05	0.15	Remainder	—
135)	63400	—	0.30 to 0.70	0.60	0.10	0.30	0.40 to 0.90	0.10	—	0.20	0.20	0.05	0.15	Remainder	—
136)	63400A	6101	0.30 to 0.70	0.50	0.10	0.03	0.35 to 0.80	0.03	—	0.10	-	0.03	0.10	Remainder	B : 0.06
137)	63400B	6401	0.35 to 0.70	0.04	0.05 to 0.20	0.03	0.35 to 0.70	—	—	0.04	0.01	0.01	-	Remainder	—
138)	63401	—	0.30 to 0.70	0.50	0.05	0.03	0.40 to 0.90	0.03	—	0.10	0.10	0.03	0.10	Remainder	—
139)	63432	6009	0.60 to 1.00	0.50	0.15 to 0.60	0.20 to 0.80	0.40 to 0.80	0.10	—	0.25	0.10	0.05	0.15	Remainder	—

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SI No.	IS Designation	ISO Designation	Si	Fe	Cu	Mn	Mg	Cr	Ni	Zn	Ti	Others		Al	Remarks
												Each	Total		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
140)	64300	6560	0.30 to 0.70	0.10 to 0.30	0.05 to 0.20	0.20	0.20 to 0.60	0.05	—	0.15	0.10	0.05	0.15	Remainder	—
141)	64423	—	0.70 to 1.30	0.80	0.50 to 1.00	1.00	0.50 to 1.30	—	—	0.20	—	—	—	Remainder	—
142)	64430	—	0.60 to 1.30	0.60	0.10	0.40 to 1.00	0.40 to 1.20	0.25	—	0.10	0.20	—	—	Remainder	—
143)	64430A	6351	0.70 to 1.30	0.50	0.10	0.40 to 0.80	0.40 to 0.80	—	—	0.20	0.20	0.05	0.15	Remainder	—
144)	64430B	6081	0.70 to 1.10	0.50	0.10	0.10 to 0.45	0.60 to 1.00	0.10	—	0.20	0.15	0.05	0.15	Remainder	—
145)	64430C	6351A	0.70 to 1.30	0.50	0.10	0.40 to 0.80	0.40 to 0.80	—	—	0.20	0.20	0.05	0.15	Remainder	Pb : 0.003
146)	64440	6951	0.20 to 0.50	0.80	0.15 to 0.40	0.10	0.40 to 0.80	—	—	0.20	—	0.05	0.15	Remainder	—
147)	65028	—	0.40 to 0.80	0.70	0.15 to 0.40	0.20	0.70 to 1.20	0.15 to 0.35	—	0.20	0.20	—	—	Remainder	—
148)	65028A	6061	0.40 to 0.80	0.70	0.15 to 0.40	0.15	0.80 to 1.20	0.04 to 0.35	—	0.25	0.15	0.05	0.15	Remainder	—
149)	65028B	6061A	0.40 to 0.80	0.70	0.15 to 0.40	0.15	0.80 to 1.20	0.04 to 0.35	—	0.25	0.15	0.05	0.15	Remainder	Pb : 0.003
150)	65028C	6064A	0.40 to 0.80	0.70	0.15 to 0.40	0.15	0.80 to 1.20	0.04 to 0.14	—	0.25	0.15	0.05	0.15	Remainder	Bi : 0.40 to 0.80 ; Pb : 0.20 to 0.40

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SI No.	IS Designation	ISO Designation	Si	Fe	Cu	Mn	Mg	Cr	Ni	Zn	Ti	Others		Al	Remarks
												Each	Total		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
151)	65032	—	0.40 to 0.80	0.70	0.15 to 0.40	0.20 to 0.80	0.70 to 1.20	0.20	—	0.20	0.20	—	—	Remainder	—
152)	65032A	6261	0.40 to 0.70	0.40	0.15 to 0.40	0.20 to 0.35	0.70 to 1.00	0.10	—	0.20	0.10	0.05	0.15	Remainder	—
153)	65423	6013	0.60 to 1.00	0.50	0.60 to 1.10	0.20 to 0.80	0.80 to 1.20	0.10	—	0.25	0.10	0.05	0.15	Remainder	—
154)	65423A	6056	0.70 to 1.30	0.50	0.50 to 1.10	0.40 to 1.00	0.60 to 1.20	0.25	—	0.10 to 0.70	—	0.05	0.15	Remainder	Zr + Ti : 0.20
155)	65430	6082	0.70 to 1.30	0.50	0.10	0.40 to 1.00	0.60 to 1.20	0.25	—	0.20	0.10	0.05	0.15	Remainder	—
156)	65430A	6082A	0.70 to 1.30	0.50	0.10	0.40 to 1.00	0.60 to 1.20	0.25	—	0.20	0.10	0.05	0.15	Remainder	Pb : 0.003
157)	66070	6070	1.00 to 1.70	0.50	0.15 to 0.40	0.40 to 1.00	0.50 to 1.20	0.10	—	0.25	0.15	0.05	0.15	Remainder	—
158)	66400	6015	0.20 to 0.40	0.10 to 0.30	0.10 to 0.25	0.10	0.80 to 1.10	0.10	—	0.10	0.10	0.05	0.15	Remainder	—
159)	66423	6066	0.90 to 1.80	0.50	0.70 to 1.20	0.60 to 1.10	0.80 to 1.40	0.40	—	0.25	0.20	0.05	0.15	Remainder	—
160)	66500	6020	0.40 to 0.90	0.50	0.30 to 0.90	0.35	0.60 to 1.20	0.15	—	0.20	0.15	0.05	0.15	Remainder	Pb : 0.05 ; Sn : 0.90 to 1.50
161)	66540	6162	0.40 to 0.80	0.50	0.20	0.10	0.70 to 1.10	0.10	—	0.25	0.10	0.05	0.15	Remainder	—

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SI No.	IS Designation	ISO Designation	Si	Fe	Cu	Mn	Mg	Cr	Ni	Zn	Ti	Others		Al	Remarks
												Each	Total		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
162)	66600	6041	0.50 to 0.90	0.15 to 0.70	0.15 to 0.60	0.05 to 0.20	0.80 to 1.20	0.05 to 0.15	—	0.25	0.15	0.05	0.15	Remainder	Bi : 0.30 to 0.90 ; Sn : 0.35 to 1.20
163)	66700	6042	0.50 to 1.20	0.70	0.20 to 0.60	0.40	0.70 to 1.20	0.04 to 0.35	—	0.25	0.15	0.05	0.15	Remainder	Bi : 0.20 to 0.80 ; Pb : 0.15 to 0.40
164)	67650	6012A	0.60 to 1.40	0.50	0.40	0.20 to 1.00	0.60 to 1.20	0.30	—	0.30	0.20	0.05	0.15	Remainder	Bi: 0.70 ; Sn: 0.40 to 2.00
165)	67660	6011	0.60 to 1.20	1.00	0.40 to 0.90	0.80	0.60 to 1.20	0.30	0.20	1.50	0.20	0.05	0.15	Remainder	—
166)	71000	7072	—	—	0.10	0.10	0.10	—	—	0.80 to 1.30	—	0.05	0.15	Remainder	Si + Fe : 0.70
167)	74530	—	0.40	0.70	0.20	0.20 to 0.70	1.00 to 1.50	0.20	—	4.00 to 5.00	0.20	—	—	Remainder	—
168)	74538	7039	0.30	0.40	0.10	0.10 to 0.40	2.30 to 3.30	0.15 to 0.25	—	3.50 to 4.50	0.10	0.05	0.15	Remainder	—
169)	75530	7005	0.35	0.40	0.10	0.20 to 0.70	1.00 to 1.80	0.06 to 0.20	—	4.00 to 5.00	0.01 to 0.06	0.05	0.15	Remainder	Zr : 0.08 to 0.20
170)	76250	7050	0.12	0.15	2.00 to 2.60	0.10	1.90 to 2.60	0.04	—	5.70 to 6.70	0.06	0.05	0.15	Remainder	Zr : 0.08 to 0.15
171)	76520	7010	0.12	0.15	1.50 to 2.00	0.10	2.10 to 2.60	0.05	0.05	5.70 to 6.70	0.06	0.05	0.15	Remainder	Zr : 0.10 to 0.16
172)	76528	7075	0.40	0.50	1.20 to 2.00	0.30	2.10 to 2.90	0.18 to 0.28	—	5.10 to 6.10	0.20	0.05	0.15	Remainder	—

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SI No.	IS Designation	ISO Designation	Si	Fe	Cu	Mn	Mg	Cr	Ni	Zn	Ti	Others		Al	Remarks
												Each	Total		
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
173)	76528A	7175	0.15	0.20	1.20 to 2.00	0.10	2.10 to 2.90	0.18 to 0.28	—	5.10 to 6.10	0.10	0.05	0.15	Remainder	—
174)	81000	8021	0.15	1.20 to 1.70	0.05	—	—	—	—	—	—	0.05	0.15	Remainder	—
175)	81000B	8021B	0.40	1.10 to 1.70	0.05	0.03	0.01	0.03	—	0.05	0.05	0.03	0.10	Remainder	—
176)	81400	8011	0.50 to 0.90	0.60 to 1.00	0.10	0.20	0.05	0.05	—	0.10	0.08	0.05	0.15	Remainder	—
177)	81400A	8011A	0.40 to 0.80	0.50 to 1.00	0.10	0.10	0.10	0.10	—	0.10	0.05	0.05	0.15	Remainder	—
178)	81400B	8111	0.30 to 1.10	0.40 to 1.00	0.10	0.10	0.05	0.05	—	0.10	0.08	0.05	0.15	Remainder	—
179)	81400C	8018	0.50 to 0.90	0.60 to 1.00	0.30 to 0.60	0.30	—	—	—	—	0.006 to 0.06	0.05	0.15	Remainder	—
180)	81472	8079	0.05 to 0.30	0.70 to 1.30	0.05	—	—	—	—	0.10	—	0.05	0.15	Remainder	—
181)	82300	8006	0.40	1.20 to 2.00	0.30	0.30 to 1.00	0.10	—	—	0.10	—	0.05	0.15	Remainder	—

NOTES

1 Aluminium shall be determined by difference. The aluminium content for unalloyed aluminium (1 series) is the difference between 100.00 percent and the sum of all other metallic elements.

2 'Others' includes the listed elements for which no specific limits are mentioned and also unlisted metallic elements. Identification of unlisted elements shall be as per the mutual agreement between the manufacturer and the purchaser. 'Others' does not include modifying or refining elements such as Na, Sr, Sb and P.

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‘(Page 9, clause 8, Table 2) — Substitute the following for the existing table.

Table 2 Mechanical Properties of Wrought Aluminium and Aluminium Alloy Sheet and Strip

(Clause 8)

SI No.	IS Designation	ISO Designation	Condition	0.2 Percent Proof Stress	Tensile Strength		Elongation on 50 mm Gauge Length, Percent, for Thickness (Min)					Bend Test, Radius of Bend
				MPa	MPa		0.2 to 0.5 mm	Over 0.5 mm, up to and Including 0.8 mm	Over 0.8 mm, up to and Including 1.3 mm	Over 1.3 mm, up to and Including 2.6 mm	Over 2.6 mm, up to 6.0 mm	
				Min	Min	Max						
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
1)	19000	—	F	—	75	—	—	—	—	—	—	—
			O	—	70	110	—	20	25	29	30	close
			H12 or H22	—	90	130	—	5	6	8	8	close
			H14 or H24	—	105	140	—	3	4	5	5	0.5t
			H16 or H26	—	125	150	—	2	3	4	4	0.5t
			H18	—	140	—	—	2	2	3	3	1t
2)	19000A	1200	O	25	75	105	17	22	22	30	30	close
			H12 or H22	75	95	125	3	4	6	8	8	0.5t
			H14 or H24	95	120	145	2	3	4	5	5	1t
			H16 or H26	115	135	165	1	2	3	4	4	2t
			H18	130	150	—	1	2	3	4	4	—
3)	19002	1100	O	25	75	105	17	22	22	30	30	close
			H12 or H22	75	95	125	3	4	6	8	8	0.5t
			H14 or H24	95	120	145	2	3	4	5	5	1t
			H16 or H26	115	135	165	1	2	3	4	4	2t
			H18 or H28	130	150	—	1	2	3	4	4	—

SI No.	IS Designation	ISO Designation	Condition	0.2 Percent Proof Stress	Tensile Strength		Elongation on 50 mm Gauge Length, Percent, for Thickness (Min)					Bend Test, Radius of Bend
				MPa	MPa		0.2 to 0.5 mm	Over 0.5 mm, up to and Including 0.8 mm	Over 0.8 mm, up to and Including 1.3 mm	Over 1.3 mm, up to and Including 2.6 mm	Over 2.6 mm, up to 6.0 mm	
				Min	Min	Max						
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
4)	19350	1235	O	25	75	105	17	22	22	30	30	close
			H12 or H22	75	95	125	3	4	6	8	8	0.5t
			H14 or H24	95	120	145	2	3	4	5	5	1t
			H16 or H26	115	135	165	1	2	3	4	4	2t
			H18 or H28	130	150	—	1	2	3	4	4	—
5)	19500	1050	F	—	65	—	—	—	—	—	—	—
			O	20	60	100	15	22	25	29	30	close
			H12 or H22	65	80	120	2	4	6	8	8	0.5t
			H14 or H24	75	100	135	1	4	5	6	6	0.5t
			H16 or H26	85	120	145	1	2	3	4	4	2t
			H18	—	125	—	1	2	3	4	4	—
6)	19500A	1050A	O	20	65	95	20	22	22	22	26	—
			H14	85	105	145	2	3	3	3	4	—
			H18	120	120	—	1	2	2	2	2	—

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SI No.	IS Designation	ISO Designation	Condition	0.2 Percent Proof Stress	Tensile Strength		Elongation on 50 mm Gauge Length, Percent, for Thickness (Min)					Bend Test, Radius of Bend
				MPa	MPa		0.2 to 0.5 mm	Over 0.5 mm, up to and Including 0.8 mm	Over 0.8 mm, up to and Including 1.3 mm	Over 1.3 mm, up to and Including 2.6 mm	Over 2.6 mm, up to 6.0 mm	
				Min	Min	Max						
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
7)	19501	–	F	–	60	–	–	–	–	–	25	1t
			H12	–	85	–	–	–	–	–	10	1t
			H14	–	95	–	–	–	–	–	6	1t
8)	19600	–	O	–	–	95	–	25	25	29	32	close
			H14 or H24	–	95	125	–	4	5	6	6	0.5t
			H18 or H28	–	125	–	–	3	3	4	4	1t
9)	19700	1070	F	–	60	–	–	–	–	–	–	–
			O	15	55	95	20	25	30	35	35	close
			H12 or H22	55	70	110	3	4	6	8	8	close
			H14 or H24	65	85	120	2	3	4	5	5	1t
			H16 or H26	75	100	135	1	2	3	4	4	1.5t
			H18 or H28	–	120	–	1	2	3	4	4	–
10)	19700A	1070A	O	15	60	90	23	25	25	25	29	0.5t
			H12	55	80	120	5	6	6	6	7	0.5t
			H14	70	100	140	4	4	4	4	5	1t
			H16	90	110	150	2	2	2	2	3	1t
			H18	105	125	–	2	2	2	2	2	–

SI No.	IS Designation	ISO Designation	Condition	0.2 Percent Proof Stress	Tensile Strength		Elongation on 50 mm Gauge Length, Percent, for Thickness (Min)					Bend Test, Radius of Bend
				MPa	MPa		0.2 to 0.5 mm	Over 0.5 mm, up to and Including 0.8 mm	Over 0.8 mm, up to and Including 1.3 mm	Over 1.3 mm, up to and Including 2.6 mm	Over 2.6 mm, up to 6.0 mm	
				Min	Min	Max						
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
11)	19800	1080	O	15	55	95	20	25	30	35	35	close
			H12 or H22	55	70	110	3	4	6	8	8	close
			H14 or H24	65	85	120	2	3	4	5	5	1t
			H16 or H26	75	100	135	1	2	3	4	4	1.5t
			H18 or H28	–	120	–	1	2	3	4	4	–
12)	19800A	1080A	O	15	60	90	26	28	28	28	31	0.5t
			H12	55	80	120	5	6	6	6	7	0.5t
			H14	70	100	140	4	4	4	4	5	1t
			H16	90	110	150	2	2	2	2	3	1t
			H18	105	125	–	2	2	2	2	2	–
13)	19850	1085	O	15	55	95	20	25	30	35	35	close
			H12	55	70	110	3	4	6	8	8	close
			H14	65	85	120	2	3	4	5	5	1t
			H16	75	100	135	1	2	3	4	4	1.5t
			H18	–	120	–	1	2	3	4	4	–
14)	19990	1199	O	–	–	65	–	30	35	40	45	close
			H14 or H24	–	80	100	–	7	6	10	12	0.5t
			H18 or H28	–	100	-	–	3	4	5	6	1t

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SI No.	IS Designation	ISO Designation	Condition	0.2 Percent Proof Stress	Tensile Strength		Elongation on 50 mm Gauge Length, Percent, for Thickness (Min)					Bend Test, Radius of Bend
				MPa	MPa		0.2 to 0.5 mm	Over 0.5 mm, up to and Including 0.8 mm	Over 0.8 mm, up to and Including 1.3 mm	Over 1.3 mm, up to and Including 2.6 mm	Over 2.6 mm, up to 6.0 mm	
				Min	Min	Max						
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
15)	24345	–	O	–	–	240	–	14	14	14	14	close
			T4	240	380	–	–	13	14	14	14	3t
			T6	345	425	–	–	6	6	6	6	5t
16)	24345A	2014	O	–	–	220	16	16	16	16	16	1.5t
			T4	240	395	–	14	14	14	14	14	3.5t
			T6	390	440	–	6	6	6	7	7	5t
17)	24345B	2014A	O	–	–	220	16	16	16	16	16	–
			T3 or T4	240	395	–	14	14	14	14	14	–
			T6	380	440	–	6	6	6	7	7	–
18)	24530	2024	O	–	–	220	12	12	12	12	12	close
			T4	275	425	–	12	15	15	15	15	3t
			T6	345	440	–	5	5	5	5	5	5t
19)	24534A	2017A	O	–	–	225	12	12	12	12	12	–
			T4	245	390	–	14	14	14	14	14	–
20)	24534B	2017	O	–	–	215	12	12	12	12	12	1.5t
			T3	215	375	–	12	15	15	17	17	3.5t
			T4	195	355	–	12	15	15	17	17	3.5t
21)	26388	2219	O	–	–	220	–	12	12	12	12	2t
			T6	250	370	–	–	6	6	7	8	5t

SI No.	IS Designation	ISO Designation	Condition	0.2 Percent Proof Stress	Tensile Strength		Elongation on 50 mm Gauge Length, Percent, for Thickness (Min)					Bend Test, Radius of Bend
				MPa	MPa		0.2 to 0.5 mm	Over 0.5 mm, up to and Including 0.8 mm	Over 0.8 mm, up to and Including 1.3 mm	Over 1.3 mm, up to and Including 2.6 mm	Over 2.6 mm, up to 6.0 mm	
				Min	Min	Max						
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
22)	31000	–	F	–	95	–	–	–	–	–	–	–
			O	–	90	130	17	20	23	24	24	Close
			H12 or H22	–	115	150	3	5	6	7	8	Close
			H14 or H24	–	130	180	1	3	4	5	5	0.5t
			H16 or H26	–	150	195	1	2	3	4	4	1t
			H18 or H28	–	170	–	1	2	2	3	3	3t
23)	31000A	3103	O	35	90	130	17	19	19	19	21	–
			H 14	120	140	180	2	2	2	2	4	–
			H18	165	185	–	1	2	2	2	2	–
24)	31200	3003	O	35	95	135	18	20	23	24	25	Close
			H12 or H22	80	120	155	3	6	7	8	9	Close
			H14 or H24	115	140	180	2	4	4	5	6	0.5t
			H16 or H26	145	165	205	1	3	3	4	4	1t
			H18	165	180	–	1	2	2	3	3	3t
25)	31500	–	O	–	125	165	10	16	16	18	20	Close
			H12 or H22	–	150	210	1	5	5	6	8	Close
			H14 or H24	–	190	245	1	3	4	5	5	0.5t
			H16 or H26	–	215	275	1	2	2	3	4	0.5t
			H18 or H28	–	245	–	1	1	1	1	2	1t

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SI No.	IS Designation	ISO Designation	Condition	0.2 Percent Proof Stress	Tensile Strength		Elongation on 50 mm Gauge Length, Percent, for Thickness (Min)					Bend Test, Radius of Bend
				MPa	MPa		0.2 to 0.5 mm	Over 0.5 mm, up to and Including 0.8 mm	Over 0.8 mm, up to and Including 1.3 mm	Over 1.3 mm, up to and Including 2.6 mm	Over 2.6 mm, up to 6.0 mm	
				Min	Min	Max						
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
26)	31500A	3004	O	60	155	195	10	14	16	18	18	0.5t
			H12 or H22 or H32	145	195	245	1	3	4	5	5	1t
			H14 or H24 or H34	175	225	265	1	3	3	4	4	1.5t
			H16 or H26 or H36	195	245	285	1	2	3	4	4	2.5t
			H18 or H28 or H36	215	265	–	1	–	–	–	–	–
			H19 or H29 or H39	–	275	–	1	–	–	–	–	–
27)	31500B	–	O	–	95	145	14	16	16	18	20	Close
			H12 or H22	–	130	180	3	5	5	6	8	0.5t
			H14 or H24	–	150	200	2	3	4	5	5	1t
			H16 or H26	–	170	220	1	2	2	3	4	2t
			H18 or H28	–	190	–	1	1	1	1	2	4t
28)	31500C	3105A	O	40	100	155	14	15	15	15	17	0.5t
			H12	105	130	180	3	4	4	4	4	1.5t
			H14	130	150	200	2	2	2	2	2	2.5t
			H16	160	175	225	1	2	2	2	2	–
			H18	180	195	–	1	1	1	1	1	–

SI No.	IS Designation	ISO Designation	Condition	0.2 Percent Proof Stress	Tensile Strength		Elongation on 50 mm Gauge Length, Percent, for Thickness (Min)					Bend Test, Radius of Bend
				MPa	MPa		0.2 to 0.5 mm	Over 0.5 mm, up to and Including 0.8 mm	Over 0.8 mm, up to and Including 1.3 mm	Over 1.3 mm, up to and Including 2.6 mm	Over 2.6 mm, up to 6.0 mm	
				Min	Min	Max						
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
29)	31540	3104	O	60	155	195	10	14	16	18	18	0.5t
			H12 or H22 or H32	145	195	245	1	3	4	5	5	1t
			H14 or H24 or H34	175	225	265	1	3	3	4	4	1.5t
			H16 or H26 or H36	195	245	285	1	2	3	4	4	2.5t
			H18 or H28 or H36	215	265	—	1	—	—	—	—	—
			H19 or H29 or H39	—	275	—	1	—	—	—	—	—
30)	31542	3005	O	45	120	165	14	16	18	18	—	0t
			H12 or H22	120	135	185	1	2	2	2	—	1t
			H14 or H24	145	165	215	1	1	2	2	—	2t
			H16 or H26	165	195	245	1	1	2	2	—	3t
			H18	205	225	—	1	1	2	2	—	—
31)	40800	—	F	—	90	—	—	—	—	—	—	—
			O	—	85	120	19	20	23	25	30	close
			H12 or H22	—	105	140	4	5	6	7	8	close
			H14 or H24	—	125	160	1	3	4	5	5	0.5t
			H16 or H26	—	150	180	1	2	3	4	4	1t
			H18 or H28	—	175	—	1	2	2	3	3	3t

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SI No.	IS Designation	ISO Designation	Condition	0.2 Percent Proof Stress	Tensile Strength		Elongation on 50 mm Gauge Length, Percent, for Thickness (Min)					Bend Test, Radius of Bend
				MPa	MPa		0.2 to 0.5 mm	Over 0.5 mm, up to and Including 0.8 mm	Over 0.8 mm, up to and Including 1.3 mm	Over 1.3 mm, up to and Including 2.6 mm	Over 2.6 mm, up to 6.0 mm	
				Min	Min	Max						
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
32)	41352	4015	O	45	–	150	20	20	20	20	20	–
			H12	90	120	175	4	4	4	4	4	–
			H14	120	150	200	2	3	3	3	3	–
			H16	150	170	220	1	2	2	2	2	–
			H18	180	200	250	1	1	1	1	1	–
33)	41352A	4017	O	45	–	150	20	20	20	20	20	–
			H12	90	120	175	4	4	4	4	4	–
			H14	120	150	200	2	3	3	3	3	–
			H16	150	170	220	1	2	2	2	2	–
34)	41800	4006	O	40	95	130	17	19	19	19	22	1t
			H12	90	120	160	4	4	4	4	5	1.5t
			H14	120	140	180	3	3	3	3	3	2t
35)	43000	–	O	–	–	130	–	–	–	–	–	–
36)	46000	–	O	–	–	150	–	–	–	–	–	–
37)	51000A	5005	O	–	105	150	–	18	18	18	22	close
			H12	85	120	160	–	3	4	5	6	close
			H14	105	140	180	–	2	3	4	4	0.5t
			H16	125	160	200	–	1	2	2	2	1t
			H18	–	185	–	–	1	2	2	2	3t

SI No.	IS Designation	ISO Designation	Condition	0.2 Percent Proof Stress	Tensile Strength		Elongation on 50 mm Gauge Length, Percent, for Thickness (Min)					Bend Test, Radius of Bend
				MPa	MPa		0.2 to 0.5 mm	Over 0.5 mm, up to and Including 0.8 mm	Over 0.8 mm, up to and Including 1.3 mm	Over 1.3 mm, up to and Including 2.6 mm	Over 2.6 mm, up to 6.0 mm	
				Min	Min	Max						
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
38)	51000B	5050	O	45	130	170	16	17	17	17	19	0.5t
			H12	130	155	195	2	2	2	2	4	–
			H14	150	175	215	2	2	2	2	3	–
			H16	170	195	235	1	2	2	2	2	–
			H18	190	220	–	1	2	2	2	2	–
39)	51300	–	O	–	95	145	–	14	14	15	16	Close
			H12 or H22	–	130	180	–	6	6	7	8	0.5t
			H14 or H24	–	150	200	–	4	4	5	6	1t
			H16 or H26	–	175	215	–	2	2	3	4	2t
			H18 or H28	–	195	–	–	1	1	1	2	4t
40)	51300A	5010	O	35	90	130	17	19	19	19	21	1t
			H12	85	110	155	2	3	3	3	4	2t
			H14	115	140	175	2	2	2	2	3	2.5t
			H16	140	155	195	1	2	2	2	2	2.5t
			H18	160	175	–	1	2	2	2	2	–

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SI No.	IS Designation	ISO Designation	Condition	0.2 Percent Proof Stress	Tensile Strength		Elongation on 50 mm Gauge Length, Percent, for Thickness (Min)					Bend Test, Radius of Bend
				MPa	MPa		0.2 to 0.5 mm	Over 0.5 mm, up to and Including 0.8 mm	Over 0.8 mm, up to and Including 1.3 mm	Over 1.3 mm, up to and Including 2.6 mm	Over 2.6 mm, up to 6.0 mm	
				Min	Min	Max						
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
41)	52000	–	O	60	175	215	13	16	16	16	18	Close
			H12 or H22	125	200	240	3	3	4	5	6	0.5t
			H14 or H24	175	230	275	2	2	2	3	4	1t
			H16 or H26	190	235	295	1	2	2	3	–	–
			H18 or H28	215	265	–	1	1	2	3	–	–
42)	52000A	5251	O	60	160	200	13	14	14	14	16	0.5t
			H12	150	190	230	3	4	4	4	5	2t
			H14	170	210	250	2	2	2	2	3	2.5t
			H16	200	230	270	1	2	2	2	3	3.5t
			H18	230	255	–	1	2	2	2	2	
43)	52300	–	O	75	160	210	–	12	14	16	18	Close
			H12 or H22	140	170	220	–	4	5	6	7	1t
			H14 or H24	180	190	250	–	4	4	5	5	2t
			H16 or H26	200	225	285	–	3	3	4	–	–
			H18 or H28	225	250	-	–	2	2	3	–	–

SI No.	IS Designation	ISO Designation	Condition	0.2 Percent Proof Stress	Tensile Strength		Elongation on 50 mm Gauge Length, Percent, for Thickness (Min)					Bend Test, Radius of Bend
				MPa	MPa		0.2 to 0.5 mm	Over 0.5 mm, up to and Including 0.8 mm	Over 0.8 mm, up to and Including 1.3 mm	Over 1.3 mm, up to and Including 2.6 mm	Over 2.6 mm, up to 6.0 mm	
				Min	Min	Max						
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
44)	52300A	–	O	–	160	210	–	12	13	14	14	Close
			H12 or H22	–	190	260	–	4	4	5	5	2t
			H14 or H24	–	220	280	–	3	3	4	4	3t
			H16 or H26	–	250	300	–	2	2	3	3	–
			H18 or H28	–	270	–	–	2	2	3	3	–
45)	52302	5049	O	80	190	240	12	14	14	14	16	1t
			H12	170	220	270	4	5	5	5	6	–
			H14	190	240	280	3	3	3	3	4	–
			H16	220	265	305	2	3	3	3	3	–
			H18	250	290	–	1	2	2	2	2	–
46)	53000	–	O	85	210	270	–	12	14	16	18	Close
			H 12	160	240	290	–	4	5	6	7	1t
			H 14	220	270	320	–	3	3	5	5	2t
			H 16	225	290	340	–	2	2	4	–	–
			H 18	235	310	–	–	2	2	3	–	–
47)	53000A	5754	O	80	190	240	12	14	14	14	16	–
			H24 or H34	160	240	280	6	6	6	6	7	–
			H38	230	290	–	3	3	3	3	4	–
48)	53380	5454	O	85	215	285	12	12	14	16	18	–

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SI No.	IS Designation	ISO Designation	Condition	0.2 Percent Proof Stress	Tensile Strength		Elongation on 50 mm Gauge Length, Percent, for Thickness (Min)					Bend Test, Radius of Bend
				MPa	MPa		0.2 to 0.5 mm	Over 0.5 mm, up to and Including 0.8 mm	Over 0.8 mm, up to and Including 1.3 mm	Over 1.3 mm, up to and Including 2.6 mm	Over 2.6 mm, up to 6.0 mm	
				Min	Min	Max						
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
49)	53800	5052	O	65	170	215	15	17	17	19	19	1t
			H12 or H22 or H32	155	215	265	4	5	5	7	7	1.5t
			H14 or H24 or H34	180	235	285	3	4	4	6	6	2t
			H16 or H26 or H36	205	255	305	3	3	4	4	4	3t
			H18 or H38	220	270	–	3	3	4	4	4	–
50)	54000	5154A	O	85	215	275	12	13	13	13	15	1t
			H12	190	250	305	3	4	4	4	5	–
			H14	220	270	325	2	3	3	3	3	–
			H18	270	310	–	1	1	1	1	1	–
51)	54000A	5154	O	75	205	285	–	12	14	16	16	1.5t
			H12 or H22 or H32	175	255	295	–	5	5	8	8	2t
			H14 or H24 or H34	205	275	315	–	4	4	6	6	2.5t
			H16 or H26 or H36	225	295	335	–	3	3	4	4	4t
			H18 or H28 or H38	240	315	–	–	3	3	4	4	–
52)	54300	–	O	130	265	365	–	12	14	16	16	1t
			H12 or H22 or H32	235	315	395	–	5	6	7	7	2t
			H14 or H24 or H34	275	355	–	–	4	4	5	5	3t

SI No.	IS Designation	ISO Designation	Condition	0.2 Percent Proof Stress	Tensile Strength		Elongation on 50 mm Gauge Length, Percent, for Thickness (Min)					Bend Test, Radius of Bend
				MPa	MPa		0.2 to 0.5 mm	Over 0.5 mm, up to and Including 0.8 mm	Over 0.8 mm, up to and Including 1.3 mm	Over 1.3 mm, up to and Including 2.6 mm	Over 2.6 mm, up to 6.0 mm	
				Min	Min	Max						
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
53)	54300A	5083	O	125	275	350	—	16	16	16	16	2t
			H12 or H22 or H32	215	305	380	—	8	8	8	8	5t
			H14 or H24 or H34	270	345	405	—	—	—	6	6	—
54)	54300B	5182	H18 or H38	—	345	—	1	—	—	—	—	—
			H19 or H39	—	365	—	1	—	—	—	—	—
55)	54380	5086	O	100	245	305	—	15	15	18	18	2.5t
			H22 or H32	195	275	325	—	6	6	8	8	3t
			H24 or H34	235	305	355	—	4	5	6	6	4t
			H26 or H36	265	325	375	—	3	4	6	6	5t
			H18 or H38	285	345	—	—	3	3	—	—	—
56)	55000	—	O	130	265	365	—	12	14	16	16	close
			H12 or H22 or H32	220	310	395	—	5	6	7	7	2t
57)	64430	—	O	—	—	175	—	14	16	16	17	Close
			T4	115	200	—	—	12	15	15	15	2t
			T6	250	295	—	—	5	5	5	6	3t
58)	65028	—	O	—	—	175	—	14	16	16	18	Close
			T4	110	200	—	—	12	15	15	15	2t
			T6	235	280	—	—	5	5	5	6	3t

Amendment No. 1 to IS 737 : 2024

SI No.	IS Designation	ISO Designation	Condition	0.2 Percent Proof Stress	Tensile Strength		Elongation on 50 mm Gauge Length, Percent, for Thickness (Min)					Bend Test, Radius of Bend
				MPa	MPa		0.2 to 0.5 mm	Over 0.5 mm, up to and Including 0.8 mm	Over 0.8 mm, up to and Including 1.3 mm	Over 1.3 mm, up to and Including 2.6 mm	Over 2.6 mm, up to 6.0 mm	
				Min	Min	Max						
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
59)	65028A	6061	O	–	–	145	–	16	16	16	16	1.5t
			T4	110	205	–	–	16	16	16	16	1.5t
			T6	245	295	–	–	10	10	10	10	3t
60)	65032	–	O	–	–	175	–	14	16	16	18	Close
			T4	110	200	–	–	12	15	15	15	2t
			T6	235	280	–	–	5	5	5	6	3t
61)	65430	6082	O	–	–	150	–	16	16	16	16	2.5 t
			T4	110	205	–	–	15	15	15	15	2t
			T6	260	310	–	–	10	10	10	10	4.5t
62)	74530	–	T4	175	280	–	–	8	9	9	10	5t
			T6	270	315	–	–	6	7	7	8	5t
63)	74538	–	O	103	227	–	–	22	22	22	22	1.5t
			T6	330	400	–	–	13	13	14	14	6t
64)	76528	7075	O	–	–	275	–	10	10	11	11	1.5t
			T6	460	525	–	–	6	6	7	7	6t
65)	81000	8021	H14	–	135	185	2	–	–	–	–	–

SI No.	IS Designation	ISO Designation	Condition	0.2 Percent Proof Stress	Tensile Strength		Elongation on 50 mm Gauge Length, Percent, for Thickness (Min)					Bend Test, Radius of Bend
				MPa	MPa		0.2 to 0.5 mm	Over 0.5 mm, up to and Including 0.8 mm	Over 0.8 mm, up to and Including 1.3 mm	Over 1.3 mm, up to and Including 2.6 mm	Over 2.6 mm, up to 6.0 mm	
				Min	Min	Max						
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
66)	81400	8011	O	—	85	120	19	20	23	25	30	close
			H12 or H22	—	105	140	4	5	6	7	8	close
			H14 or H24	—	125	160	1	3	4	5	5	0.5t
			H16 or H26	—	150	180	1	2	3	4	4	1t
			H18 or H28	—	175	—	1	2	2	3	3	3t
67)	81400A	8011A	O	30	85	130	19	21	21	21	24	-
			H14	110	125	165	1	3	3	3	3	-
			H16	130	140	190	1	2	2	2	3	—
			H18	145	160	—	1	2	2	2	2	—
68)	81472	8079	H14	—	125	175	2	—	—	—	—	—
NOTES												
1 1 MPa = 1 N/mm ² = 1 MN/m ² = 0.102 kgf/mm ² = 144.4 psi.												
2 ‘t’ is the thickness of the test piece.												
3 For thickness 2.6 mm and less, elongation values are for guidance only and not guaranteed. For this purpose, bend test as specified in 8.2 may be carried out.												
4 Properties in F temper are only typical values and are given for information only.												

(MTD 07)

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